CURRICULUM VITAE

Muhammad Abu Bakar

Geospatial Data Specialist



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Professional experience

11/2025 – present

Geospatial Data Specialist · Beagle Systems, Hamburg

Beagle Systems flies long-range drones beyond visual line of sight (BVLOS) to monitor pipelines, utilities and other infrastructure.

- Built an open-source drone LiDAR pipeline that georeferences and classifies point clouds to about 5 cm vertical accuracy without ground control points.
- Automated vegetation-clearance analysis for power lines, assigning risk categories to vegetation near the conductors; now developing tree-fall risk analysis.
- Fine-tuned an RF-DETR foundation model to detect construction activity along pipelines, with a Qwen 3.5 VLM as gatekeeper that verifies every detection by majority vote.
- Automated the right-of-way workflow from deep-feature-matching georeferencing to PostGIS delivery: a full flight (~900 ha of corridor) in about 3 h on one GPU.
- Extended the monitoring with real-time heavy-machinery detection on an NVIDIA Jetson, sending results straight to the customer.
- Fine-tuned a building-segmentation foundation model and ran it on 10 cm orthophotos of the Cologne area.
- Built PostGIS and TiTiler services to serve large GeoTIFFs.
- Ran pre-flight primary and duplicate checks; helped extend the flight-planning geodatabase.

03/2024 – 10/2025

Geospatial (EO) Data Scientist, part-time · Karuna Technology UG, Germany

- Differential InSAR pipeline with ISCE: automated acquisition, preprocessing and interferogram generation.
- Google Earth Engine workflows for Sentinel-1 time series (Refined Lee filter) feeding a soil-moisture ML model.
- Urban tree-species classification from fused multispectral and SAR data; training data for tree-crown segmentation.
- Vector data pre- and post-processing in QGIS.

04/2021 – 02/2024

GIS & RS Specialist · MapTransfer, Germany

- Spatial data workflows for a construction company: georeferencing, orthophoto mosaics, cloud-optimized GeoTIFFs on AWS, PostGIS–QGIS integration.
- In-house courses on LiDAR in QGIS and spatial data analysis; course development for Spatial SQL, ALKIS and GeoServer.
- Geodata management and visualization for a telecom project based on ALKIS.
- Web GIS: PostGIS vector layers in OpenLayers.

Education

2017 – 2020

MSc Geology · University of Freiburg, Germany

- Thesis: Simulation of multispectral and hyperspectral imaging techniques for mapping of epithermal ore deposits in vegetated zones.

2016 – 2020

MSc Remote Sensing & Geographic Information System · National University of Sciences and Technology, Pakistan

- Thesis: Development of an open-source Python-based slope stability model that incorporates remote sensing and geotechnical data for slope failure assessment.

2012 – 2015

BSc Geology · Bahria University, Pakistan

- Thesis: Petrophysical analysis of Joya Mair – Well 4.

Academic projects

Landsat · Sentinel-2	Glacier mass balance of Mýrdalsjökull from historical imagery and DEMs: the melting rate doubled after 2000.
MODIS · TRMM	Snow cover and precipitation variability, 2000–2015.
Landsat 7	Snow-cover change in the Kabul Basin and its impact on water resources.
Landsat 8 · ASTER	Land cover, erosion and morphometry of Mauna Kea.

Further training

2022

NASA Applied Remote Sensing Training (ARSET): five trainings on SAR-based disaster assessment, flood monitoring and modeling, crop mapping with polarimetric SAR and optical data, air-quality model output, and pre- and post-fire monitoring.

UNAVCO / GAGE short course: InSAR processing and time-series analysis for geophysical applications (ISCE, ARIA-tools, MintPy).

Skills

GIS	ArcGIS Pro, QGIS, ArcMap
Remote sensing	Multispectral, hyperspectral and SAR data; ENVI, ERDAS Imagine, eCognition, ESA SNAP
LiDAR	CloudCompare, WhiteboxTools; DEM and DSM generation
Cloud	Google Earth Engine, AWS (EC2, S3)
Databases	PostgreSQL/PostGIS, SpatiaLite, DuckDB, SQL
Programming	Python (GDAL, Fiona, Rasterio, scikit-learn, TensorFlow, Cartopy), R, Spatial SQL, Bash
Machine learning	CatBoost, CNNs, fine-tuned foundation models, VLMs
Web mapping	Leaflet, OpenLayers, GeoServer, TiTiler
Other	Git, LaTeX, Microsoft Office

Languages

English	C2
German	B2

Freiburg im Breisgau, 14 October 2026

Signature